

Impulseantwort

1. $\psi_0(x) = \frac{1}{\sqrt{\sqrt{\pi} x_0}} \exp\left(-\frac{x^2}{2x_0^2}\right) = \langle x | \psi \rangle$

$$\langle p | \psi \rangle = \langle p | \left(\int dx |x\rangle \langle x| \right) | \psi \rangle = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} \psi(x) e^{-\frac{ipx}{\hbar}} dx$$

$$\begin{aligned} \psi(p) &= \frac{1}{\sqrt{2\pi\hbar}} \frac{1}{\pi^{1/4} x_0} \int_{-\infty}^{+\infty} e^{-\frac{x^2}{2x_0^2} - \frac{ipx}{\hbar}} dx = \frac{\sqrt{x_0}}{\pi^{1/4} \sqrt{\hbar}} e^{-\frac{p^2 x_0^2}{2\hbar^2}} = \\ &= \frac{1}{\pi^{1/4} \sqrt{p_0}} e^{-\frac{p^2}{2p_0^2}} \Rightarrow w(p) = |\psi(p)|^2 = \frac{1}{\sqrt{\pi} p_0} e^{-\frac{p^2}{p_0^2}} \end{aligned}$$

3. $\hat{x} = x_0 \frac{\hat{a} + \hat{a}^\dagger}{\sqrt{2}} \quad x_0 = \sqrt{\frac{\hbar}{m\omega}}$

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$$

$$\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$$

1) $\langle m | \hat{x} | n \rangle = \frac{x_0}{\sqrt{2}} [\langle m | \hat{a} | n \rangle + \langle m | \hat{a}^\dagger | n \rangle]$

$$\langle m | \hat{a} | n \rangle = \sqrt{n} \langle m | n-1 \rangle = \sqrt{n} \delta_{m,n-1}$$

$$\langle m | \hat{a}^\dagger | n \rangle = \sqrt{n+1} \langle m | n+1 \rangle = \sqrt{n+1} \delta_{m,n+1}$$

$$\langle m | \hat{x} | n \rangle = \frac{x_0}{\sqrt{2}} (\sqrt{n} \delta_{m,n-1} + \sqrt{n+1} \delta_{m,n+1})$$

2) $\hat{x}^2 = \frac{x_0^2}{2} (\hat{a} + \hat{a}^\dagger)^2 = \frac{x_0^2}{2} (\hat{a}^2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} + (\hat{a}^\dagger)^2)$

$$\hat{x}^2|n\rangle = \frac{x_0^2}{2} [\hat{a}^2|n\rangle + \hat{a}\hat{a}^\dagger|n\rangle + \hat{a}^\dagger\hat{a}|n\rangle + (\hat{a}^\dagger)^2|n\rangle] =$$

$$= \frac{x_0^2}{2} [\sqrt{n(n-1)}|n-2\rangle + (2n+1)|n\rangle + \sqrt{(n+1)(n+2)}|n+2\rangle] \Rightarrow$$

$$\Rightarrow \langle m | \hat{x}^2 | n \rangle = \frac{x_0^2}{2} [\sqrt{n(n-1)} \delta_{m,n-2} + (2n+1) \delta_{m,n} + \sqrt{(n+1)(n+2)} \delta_{m,n+2}]$$

3) $\hat{x}^3 = \left(\frac{x_0}{\sqrt{2}}\right)^3 (\hat{a} + \hat{a}^\dagger)^3 = \frac{x_0}{\sqrt{2}} (\hat{a} + \hat{a}^\dagger) \frac{x_0^2}{2} [\sqrt{n(n-1)}|n-2\rangle +$

$$+ (2n+1)|n\rangle + \sqrt{(n+1)(n+2)}|n+2\rangle] \Rightarrow$$

$$\Rightarrow \langle m | \hat{x}^3 | n \rangle = \frac{x_0^3}{2\sqrt{2}} \left[\sqrt{n(n-1)(n-2)} \delta_{m,n-2} + 3n^{3/2} \delta_{m,n-1} + \right. \\ \left. + 3(n+1)^{3/2} \delta_{m,n+1} + \sqrt{(n+1)(n+2)(n+3)} \delta_{m,n+3} \right]$$

5. $\hat{L}_i = \frac{1}{\hbar} \varepsilon_{ijk} \hat{x}_j \hat{p}_k$

1) $[\hat{L}_i, \hat{x}_j] = \frac{1}{\hbar} [\hat{x}_e \hat{p}_m, x_j] \varepsilon_{iem} = \frac{1}{\hbar} \varepsilon_{iem} (x_e [\hat{p}_m, x_j] + \underbrace{[x_e, x_j] \hat{p}_m}_0) = i \varepsilon_{ije} \hat{x}_e$

2) $[\hat{L}_i, \hat{p}_j] = i \varepsilon_{ije} \hat{x}_e$

3) $[\hat{L}_i, \hat{x}_j \hat{p}_j] = [\hat{L}_i, \hat{x}_j] \hat{p}_j + \hat{x}_j [\hat{L}_i, \hat{p}_j] =$
 $= i \underbrace{\varepsilon_{ijk}}_{\text{antisymm}} (\underbrace{\hat{x}_k \hat{p}_j + \hat{x}_j \hat{p}_k}_{\text{symm}}) = 0$

4) $\varepsilon_{ijk} \hat{L}_k = \frac{1}{\hbar} \varepsilon_{ijk} \varepsilon_{emn} \hat{x}_m \hat{p}_n = \frac{1}{\hbar} (\hat{x}_i \hat{p}_j - \hat{x}_j \hat{p}_i)$

$[\hat{L}_i, \hat{L}_j] = \frac{1}{\hbar^2} \varepsilon_{imn} \varepsilon_{ijk} [\hat{x}_m \hat{p}_n, \hat{x}_e \hat{p}_k] = \frac{1}{\hbar^2} \varepsilon_{imn} \varepsilon_{ijk} (\hat{x}_m \underbrace{[\hat{p}_n, \hat{x}_e] \hat{p}_k}_{-i\hbar \delta_{ne}} + \dots + \hat{x}_e [\hat{x}_m, \hat{p}_k] \hat{p}_n) = \frac{i}{\hbar} (\hat{x}_i \hat{p}_j - \hat{x}_j \hat{p}_i) = i \varepsilon_{ijk} \hat{L}_k$

5) $[\hat{L}_i, \hat{L}_j \hat{L}_j] = [\hat{L}_i, \hat{L}_j] \hat{L}_j + \hat{L}_j [\hat{L}_i, \hat{L}_j] =$
 $= i \varepsilon_{ijk} \hat{L}_k \hat{L}_j + i \varepsilon_{ijk} \hat{L}_j \hat{L}_k = 0$

6) $[\hat{L}_i, \hat{x}_j \hat{x}_j] = 0 \quad [\hat{L}_i, \hat{p}_j \hat{p}_j] = 0$

7) $[\hat{L}_i, \hat{p}_j \hat{p}_k] = [\hat{L}_i, \hat{p}_j] \hat{p}_k + \hat{p}_j [\hat{L}_i, \hat{p}_k] = i(\varepsilon_{ijm} \hat{p}_m \hat{p}_k + \varepsilon_{ikm} \hat{p}_j \hat{p}_m)$

8) Рассмотрим $F(r)$ no зависимости r : $[\hat{L}_i, r^2] = [\hat{L}_i, \hat{x}_j \hat{x}_j] = 0$
 $\Rightarrow [\hat{L}_i, r^n] = 0 \Rightarrow [\hat{L}_i, F(r)] = 0$

$$6. a) \hat{b}_i \hat{b}_j = \delta_{ij} \hat{1} + i \varepsilon_{ijk} \hat{b}_k = \frac{1}{2}(\hat{b}_i \hat{b}_j + \hat{b}_j \hat{b}_i) + \frac{1}{2}(\hat{b}_i \hat{b}_j - \hat{b}_j \hat{b}_i)$$

$$\hat{1}, \hat{b}_i, \hat{b}_j, \hat{b}_k - \text{операторы}$$

$$\hat{b}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \hat{b}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \hat{b}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$b) \text{ Пусть } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = a_0 \cdot \hat{1} + a_1 \hat{b}_1; \quad A = \begin{pmatrix} a_0 + a_3 & a_1 - i a_2 \\ a_1 + i a_2 & a_0 - a_3 \end{pmatrix} \Rightarrow$$

$$\Rightarrow a_0 = \frac{a+d}{2} \quad a_3 = \frac{a-d}{2} \quad a_1 = \frac{b+c}{2} \quad a_2 = \frac{c-b}{2}$$

$$\text{Tr} A = 2a_0 \Rightarrow a_0 = \frac{1}{2} \text{Tr}(\hat{A})$$

$$\text{Tr}(A \hat{b}_m) = a_0 \text{Tr}(\hat{b}_m) + a_1 \text{Tr}(\hat{b}_1 \hat{b}_m) = a_1 (2\delta_{1m} - 2a_m) \Rightarrow$$

$$\Rightarrow a_m = \frac{1}{2} \text{Tr}(\hat{A} \hat{b}_m)$$

7.

Проверить, что

$$a) e^{\frac{i\alpha}{2} (\vec{n} \cdot \vec{\sigma})} = \hat{1} \cos \frac{\alpha}{2} + i (\vec{n} \cdot \vec{\sigma}) \sin \frac{\alpha}{2};$$

$$b) e^{\frac{i\alpha}{2} \sigma_z} \sigma_x e^{-\frac{i\alpha}{2} \sigma_z} = \sigma_x \cos \alpha - \sigma_y \sin \alpha;$$

$$в) e^{\frac{i\alpha}{2} \sigma_z} \sigma_y e^{-\frac{i\alpha}{2} \sigma_z} = \sigma_x \sin \alpha + \sigma_y \cos \alpha,$$

$$a) (\vec{n}, \vec{b})^2 = n_i b_j n_j b_i = n_i n_j (\delta_{ij} \hat{1} + i \varepsilon_{ijk} \hat{b}_k) = \hat{1}$$

$$e^{i \frac{\alpha}{2} (\vec{n}, \vec{b})} = 1 + \frac{i\alpha}{2} (\vec{n}, \vec{b}) + \dots = \hat{1} \cos \frac{\alpha}{2} + i (\vec{n}, \vec{b}) \sin \frac{\alpha}{2}$$

$$b) (\hat{1} \cos \frac{\alpha}{2} + i \hat{b}_3 \sin \frac{\alpha}{2}) \hat{b}_x (\hat{1} \cos \frac{\alpha}{2} + i \hat{b}_3 \sin \frac{\alpha}{2}) =$$

$$= \hat{b}_x \cos^2 \frac{\alpha}{2} + \hat{b}_3 \hat{b}_x \hat{b}_3 \sin^2 \frac{\alpha}{2} + i \sin \frac{\alpha}{2} \cos \frac{\alpha}{2} \hat{b}_x \hat{b}_3 = \hat{b}_x \cos \alpha - \hat{b}_y \sin \alpha$$

$$\hat{b}_3 \hat{b}_x - \hat{b}_x \hat{b}_3 = 2i \varepsilon_{312} \hat{b}_y = 2i \hat{b}_y$$

$$\hat{b}_3 \hat{b}_x \hat{b}_3 = -i \hat{b}_3 \hat{b}_y = -i (\hat{b}_{32} + i \varepsilon_{321} \hat{b}_1) = -\hat{b}_x$$

$$8. a) \text{ Проверить } [\hat{J}_i, \hat{A}_j] = i \varepsilon_{ijk} \hat{A}_k$$

$$\hat{R} \psi = e^{i\varphi(\vec{r}, \vec{J})}$$

$$\hat{R} \psi = \hat{1} + i \delta \varphi(\vec{r}, \vec{J})$$

$$\langle \psi' | \hat{A} | \psi' \rangle = \bar{A}' \Rightarrow \vec{r}' = \vec{r} - \delta \varphi [\vec{r} \times \vec{r}] \Rightarrow$$

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$$\Rightarrow \bar{A}' = \bar{A} - \delta\varphi [\bar{n} \times \bar{A}] \Rightarrow \langle \psi' | \hat{A} | \psi' \rangle = \langle \psi | \bar{A} - \delta\varphi [\bar{n} \times \bar{A}] | \psi \rangle =$$

$$= \langle \psi | \hat{R}_{\delta\varphi}^\dagger \hat{A} \hat{R}_{\delta\varphi} | \psi \rangle \Rightarrow \hat{R}_{\delta\varphi}^\dagger \hat{A} \hat{R}_{\delta\varphi} = \hat{A} - \delta\varphi [\bar{n} \times \bar{A}] \Rightarrow$$

$$\Rightarrow (\hat{1} - i\delta\varphi(\bar{n} \cdot \vec{J})) \hat{A} (\hat{1} + i\delta\varphi(\bar{n} \cdot \vec{J})) = \hat{A} + i\delta\varphi \hat{A}(\bar{n}, \vec{J}) - (\bar{n}, \vec{J})\hat{A} \Rightarrow$$

$$\Rightarrow [\hat{J}_i, \hat{A}_j] = i\epsilon_{ijk} \hat{A}_k$$

$$\delta) [\hat{J}^2, \hat{A}_i] = [\hat{J}_j \hat{J}_j, \hat{A}_i] = J_j [J_j, A_i] + [J_j, A_i] J_j =$$

$$= i\epsilon_{ijk} \{ \hat{J}_j, \hat{A}_k \}$$

9. $\hat{H}(\lambda) |E(\lambda)\rangle = E(\lambda) |E(\lambda)\rangle \Rightarrow E_n = \langle E_n(\lambda) | \hat{H}(\lambda) | E_n(\lambda) \rangle$

$$\Rightarrow \frac{\partial \langle E_n |}{\partial \lambda} \hat{H}(\lambda) | E_n(\lambda) \rangle + \langle E_n | \frac{\partial \hat{H}(\lambda)}{\partial \lambda} | E_n \rangle + \langle E_n | \hat{H}(\lambda) \frac{\partial | E_n \rangle}{\partial \lambda} =$$

$$= \frac{\partial \langle E_n |}{\partial \lambda} E(\lambda) | E_n(\lambda) \rangle + \langle E_n | \frac{\partial E(\lambda)}{\partial \lambda} | E_n \rangle + \langle E_n | \hat{E}(\lambda) \frac{\partial | E_n \rangle}{\partial \lambda} \Rightarrow$$

$$\Rightarrow \frac{\partial E_n}{\partial \lambda} \cancel{\langle E_n | E_n \rangle} = \langle E_n | \frac{\partial \hat{H}}{\partial \lambda} | E_n \rangle$$

10. $\vec{J} = \vec{L} + \vec{S} \quad (\vec{L}, \vec{S}) = \frac{1}{2}(\vec{J}^2 - \vec{L}^2 - \vec{S}^2)$ (see on. quer.)
 $|j, j_z, l, s\rangle - \text{corr.}$

$$\bullet \langle L, S \rangle = \frac{1}{2} [\langle \hat{J}^2 \rangle - \langle \vec{L}^2 \rangle - \langle \vec{S}^2 \rangle] = \frac{1}{2} (j(j+1) - l(l+1) - s(s+1))$$

$$(\vec{J}, \vec{L}) = (\vec{L} + \vec{S}, \vec{L}) = \vec{L}^2 + (\vec{S}, \vec{L})$$

$$\langle \vec{J}, \vec{L} \rangle = l(l+1) + \frac{1}{2}(\dots) = \frac{1}{2} (j(j+1) + l(l+1) - s(s+1))$$

$$(\vec{J}, \vec{S}) = (\vec{L} + \vec{S}, \vec{S}) = (\vec{L}, \vec{S}) + \vec{S}^2 \Rightarrow \langle \vec{J}, \vec{S} \rangle = \frac{1}{2} (j(j+1) - l(l+1) + s(s+1))$$

11. $|m_1, m_2\rangle = |S_1=1, m_1\rangle \otimes |S_2=1, m_2\rangle, \quad m_i = \pm 1, 0$

$$\vec{S} = \vec{S}^{(1)} + \vec{S}^{(2)} \quad S_z = S_z^{(1)} + S_z^{(2)}$$

$$S_{\pm}^{(i)} |1, m^{(i)}\rangle = \sqrt{2 - m^{(i)}(m^{(i)} \pm 1)} |1, m^{(i)} \pm 1\rangle$$

$$\begin{aligned} 1) \quad S=1 \Rightarrow \quad S_+ |1, 1\rangle &= 0 & S_- |1, 1\rangle &= \hbar \sqrt{2} |1, 0\rangle \\ S_+ |1, 0\rangle &= \sqrt{2} |1, 1\rangle & S_- |1, 0\rangle &= \hbar \sqrt{2} |1, -1\rangle \\ S_+ |1, -1\rangle &= \sqrt{2} |1, 0\rangle & S_- |1, -1\rangle &= 0 \end{aligned}$$

$$2) \quad S=2, M=2 \Rightarrow \quad S_- |2, 2\rangle = (S_-^{(1)} + S_-^{(2)}) |1, 1\rangle =$$

$$= \sqrt{2} (|0, 1\rangle + |1, 0\rangle) ; \quad \sqrt{S(S+1) - M(M-1)} = 2 \Rightarrow$$

$$\Rightarrow |2, 1\rangle = \frac{1}{\sqrt{2}} (|0, 1\rangle + |1, 0\rangle)$$

$$3) \quad S=2, M=1 \Rightarrow \quad S_- |2, 1\rangle = \frac{1}{\sqrt{2}} (S_-^{(1)} + S_-^{(2)}) (|0, 1\rangle + |1, 0\rangle) =$$

$$= |-1, 1\rangle + |1, -1\rangle + 2|0, 0\rangle$$

$$S_- |2, 1\rangle = \sqrt{6} |2, 0\rangle \Rightarrow |2, 0\rangle = \frac{1}{\sqrt{6}} (|-1, 1\rangle + |1, -1\rangle + 2|0, 0\rangle)$$

$$4) \quad \text{by norm. } |2, -1\rangle = \frac{1}{\sqrt{2}} (|0, -1\rangle + |-1, 0\rangle)$$

flanagan const. $|1, 1\rangle$ op. $|2, 1\rangle$ c $M=1$

$$S_- |1, 1\rangle = \frac{1}{\sqrt{2}} (S_-^{(1)} + S_-^{(2)}) (|0, 1\rangle - |1, 0\rangle) = \sqrt{2} |1, 0\rangle$$

$$|1, 0\rangle = \frac{1}{\sqrt{2}} (|-1, 1\rangle - |1, -1\rangle)$$

$$\text{then } S=1, M=-1 : |1, -1\rangle = \frac{1}{\sqrt{2}} (|0, -1\rangle - |-1, 0\rangle)$$

$$5) \quad S=0 : |0, 0\rangle = \frac{1}{\sqrt{2}} (|-1, 1\rangle + |1, -1\rangle),$$

$$\text{op. } |2, 0\rangle, |1, 0\rangle \Rightarrow |0, 0\rangle = \frac{1}{\sqrt{2}} (|-1, 1\rangle - |0, 0\rangle + |1, -1\rangle)$$

Задача

1. $\hat{H} = \frac{\hat{p}_1^2}{2m} + \frac{m\omega^2 \hat{x}_1^2}{2} + \frac{\hat{p}_2^2}{2m} + \frac{m\omega^2 \hat{x}_2^2}{2} + \alpha \hat{x}_1 \hat{x}_2$

Нормы $x_1 = \frac{1}{\sqrt{2}}(\hat{x}_1 + \hat{x}_2)$, $x_2 = \frac{1}{\sqrt{2}}(\hat{x}_1 - \hat{x}_2)$

$p_1 = \frac{1}{\sqrt{2}}(\hat{p}_1 + \hat{p}_2)$, $p_2 = \frac{1}{\sqrt{2}}(\hat{p}_1 - \hat{p}_2)$

Тогда $\hat{H} = \left(\frac{p_1^2}{2m} + \frac{1}{2} m \Omega_1^2 x_1^2 \right) + \left(\frac{p_2^2}{2m} + \frac{1}{2} m \Omega_2^2 x_2^2 \right)$

$\Omega_1^2 = \omega^2 + \frac{\alpha}{m}$ $\Omega_2^2 = \omega^2 - \frac{\alpha}{m}$

$E_{n_1, n_2} = \hbar \Omega_1 (n_1 + \frac{1}{2}) + \hbar \Omega_2 (n_2 + \frac{1}{2})$

2. $-\frac{\hbar^2}{2m} \psi''(x) + U(x) \psi(x) = E \psi(x)$

$U(x) = -\frac{\hbar^2}{2m} \sum_{n=-\infty}^{+\infty} \delta(x-na)$ $k = \sqrt{\frac{-2mE}{\hbar^2}}$

$U(x) = \sum_{n=-\infty}^{+\infty} V_n e^{i \frac{2\pi n}{a} x}$ $V_n = \frac{1}{a} \int_0^a U(x) e^{-i \frac{2\pi n}{a} x} dx = -\frac{\hbar^2}{ma}$

$\psi_k(x) = e^{ikx} U_k(x)$, $U_k(x+a) = U_k(x)$

$U_k(x) = \sum_{n=-\infty}^{+\infty} C_n e^{i \frac{2\pi n}{a} x}$ $\psi_k(x) = \sum_{n=-\infty}^{+\infty} C_n e^{i(k + \frac{2\pi n}{a})x}$

$\sum_{n=-\infty}^{+\infty} \left(\frac{\hbar^2}{2m} (k + \frac{2\pi n}{a})^2 C_n + \sum_m V_{n-m} C_m \right) \cdot e^{i(k + \frac{2\pi n}{a})x} = E \sum_n C_n e^{i(k + \frac{2\pi n}{a})x} \Rightarrow$

$\Rightarrow C_n = \frac{\frac{\hbar^2}{2m} S}{\frac{\hbar^2}{2m} (k + \frac{2\pi n}{a})^2 - E}$

3. а) $\langle l, l_z | \hat{L}_x | l, l_z \rangle = \frac{1}{2} \langle l, l_z | \hat{L}_+ | l, l_z \rangle + \frac{1}{2} \langle l, l_z | \hat{L}_- | l, l_z \rangle = 0$

б) $\hat{L}_{\pm} = \hat{L}_x \pm i \hat{L}_y$, $\langle \hat{L}_{\pm}^2 \rangle = 0$

$$\text{Top } \langle \hat{L}_x^2 \rangle = \langle \hat{L}_y^2 \rangle, \quad \langle \hat{L}_x \hat{L}_y \rangle + \langle \hat{L}_y \hat{L}_x \rangle = 0$$

$$[\hat{L}_i, \hat{L}_j] = i \varepsilon_{ijk} \hat{L}_k \Rightarrow [\hat{L}_x, \hat{L}_y] = i \hat{L}_z \Rightarrow \langle [\hat{L}_x, \hat{L}_y] \rangle = i \langle \hat{L}_z \rangle$$

$$b) \underbrace{\langle \hat{L}^2 \rangle}_{l(l+1)} = \langle \hat{L}_x^2 \rangle + \langle \hat{L}_y^2 \rangle + \langle \hat{L}_z^2 \rangle \Rightarrow \langle \hat{L}_x^2 \rangle = \frac{1}{2}(l(l+1) - \langle \hat{L}_z^2 \rangle)$$

$$4. \quad (\hat{\vec{S}}, \vec{n}) = \frac{1}{2}(\vec{\sigma}, \vec{n})$$

$$(\vec{\sigma}, \vec{n}) = \sigma_x \sin \theta \cos \varphi + \sigma_y \sin \theta \sin \varphi + \sigma_z \cos \theta$$

$$(\vec{\sigma}, \vec{n}) = \begin{pmatrix} \cos \theta & \sin \theta e^{-i\varphi} \\ \sin \theta e^{i\varphi} & -\cos \theta \end{pmatrix} \quad \text{e.g. } -1 \Rightarrow$$

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$$|n_+\rangle = \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{i\varphi} \end{pmatrix} \quad |n_-\rangle = \begin{pmatrix} -\sin \frac{\theta}{2} \\ \cos \frac{\theta}{2} e^{i\varphi} \end{pmatrix}$$

$$|+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$5. \quad a) \quad w(S_z = 1/2 \mid S_a = -1/2)$$

$$|a_-\rangle = \begin{pmatrix} -\sin \frac{\theta_a}{2} \\ \cos \frac{\theta_a}{2} e^{i\varphi_a} \end{pmatrix} \quad |\psi\rangle = \sum_i c_i |\psi_i\rangle$$

$$|a_-\rangle = c_+ |+\rangle + c_- |-\rangle$$

$$w(S_z = 1/2 \mid S_a = -1/2) = |c_+|^2 = |\langle + | a_- \rangle|^2 = \sin^2 \frac{\theta_a}{2}$$

$$b) \quad w(S_a = 1/2 \mid S_b = 1/2) ; \quad \frac{1}{2}(1 + \cos(\theta_a - \theta_b)) = \frac{1}{2}(1 + (\vec{a}, \vec{b}))$$

$$\varphi_a = \varphi_b = \varphi \Rightarrow w(S_a = 1/2 \mid S_b = 1/2) = |\langle b_+ | a_+ \rangle|^2 = \cos^2 \frac{\theta}{2}$$

$$b) \quad w(S_a | S_b) = \frac{1}{2}(1 + 4 S_a S_b(\varphi, \varphi))$$

$$6. \hat{L}_z = -i \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) \quad \hat{L}_{\pm} = \hat{L}_x \pm i \hat{L}_y$$

$$\hat{L}_z = -i \frac{\partial}{\partial \phi} \quad L_{\pm} = e^{\pm i \phi} \left(\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right)$$

$$\hat{L}_x = i \left(\sin \phi \frac{\partial}{\partial \theta} + \cot \theta \sin \phi \frac{\partial}{\partial \phi} \right)$$

$$\hat{L}_y = i \left(-\cos \phi \frac{\partial}{\partial \theta} + \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right)$$

8. gW b cp. Koopa: $-\frac{\hbar^2}{2m} \left(\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{\Delta \phi}{r^2} \right) \psi +$

$$+ V(r) \psi = E \psi ; \quad \psi(r, \theta, \phi) = \underbrace{\langle r, \theta, \phi | \psi \rangle}_{|R\rangle |l, m\rangle} = \langle r | R \rangle \cdot \langle \theta, \phi | l, m \rangle$$

$$-\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R}{\partial r} \right) + \left(\frac{\hbar^2 l(l+1)}{2m r^2} + V(r) \right) R = E R$$

$$R = \frac{X(r)}{r} \Rightarrow \frac{1}{r^2} \cdot \frac{\partial}{\partial r} \left(r^2 \frac{\partial R}{\partial r} \right) = \frac{X''}{r} \Rightarrow$$

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$$\Rightarrow \frac{\hbar^2}{2m} X'' + \left(\frac{\hbar^2 l(l+1)}{2m r^2} + V(r) \right) X = E X \Rightarrow$$

$$\Rightarrow E_n = \frac{\hbar^2}{2m a^2} \frac{\pi^2 n^2}{(X_{l,m})^2} \quad \psi = \frac{1}{\sqrt{2\pi a}} \frac{\sin\left(\frac{\pi n r}{a}\right)}{r}$$

9. Bogorod: $\hat{H} = \frac{\hat{p}^2}{2} - \frac{1}{r}$

a) Dev. coor. $n=1, l=0, m=0 \quad \psi_{100}(r) = \frac{1}{\sqrt{\pi}} e^{-r}$

$$\langle \frac{1}{r} \rangle = \int_{-\infty}^{+\infty} \frac{1}{r} |\psi_{100}|^2 d^3r = 4\pi \frac{1}{\pi} \int_0^{\infty} r e^{-2r} dr = 1 \text{ a.e.} = \frac{1}{a_0}$$

$$\langle r^2 \rangle = \frac{1}{\pi} \int_0^{\infty} r^4 e^{-2r} dr \cdot 4\pi = 4 \int_0^{\infty} e^{-2r} r^4 dr = 3 = 3a_0^2$$

$$\langle \hat{p}^2 \rangle = \langle 2(\hat{H} + \frac{1}{r}) \rangle = 2(E_1 + \langle \frac{1}{r} \rangle) = 2(-\frac{1}{2} + 1) = 1 = \left(\frac{\hbar}{a_0}\right)^2$$

b) $\langle \vec{r} | \hat{H}_0(\vec{r}, l) | E_n, l, l_z \rangle = E_n(\vec{r}, l) R_{n,l}(r)$

$$\hat{H}_r(r, l) = -\frac{\hbar^2}{2m} \Delta_r + \frac{\hbar^2 l(l+1)}{2mr^2} - \frac{\gamma l^2}{r} \quad E_n(r, l) = -\frac{l^2}{2a_0} \frac{\gamma^2}{n}$$

$$\langle E_n, l, l_z | \frac{1}{r} | E_n, l, l_z \rangle = -\frac{1}{l^2} \langle E_n, l, l_z | \frac{\partial \hat{H}_r}{\partial \gamma} | E_n, l, l_z \rangle$$

$$\langle \frac{1}{r} \rangle = \frac{1}{r^2} \frac{\partial E_r}{\partial \gamma} = \frac{\gamma}{a_0 n^2} \quad , \quad \langle \frac{1}{r^2} \rangle = \frac{\gamma}{a_0^2 n^3 (l+1/2)}$$

$$b) \quad \hat{p}_r = -i \left(\frac{\partial}{\partial r} + \frac{1}{r} \right) \quad \frac{d}{dt} \langle \hat{p}_r \rangle = \langle \frac{\partial \hat{p}_r}{\partial t} \rangle + i \langle [\hat{H}, \hat{p}_r] \rangle$$

$$B \text{ const. coeff.} \quad \frac{d}{dt} \langle \hat{p}_r \rangle = 0 \quad \hat{H} = \frac{p_r^2}{2} + \frac{\hat{L}^2}{2r^2} - \frac{\gamma}{r}$$

$$[\hat{H}, \hat{p}_r] = i \frac{\hat{L}^2}{r^3} + i \frac{\gamma}{r^2} \quad ; \quad \langle [\hat{H}, \hat{p}_r] \rangle = 0 = i \left(\langle \frac{\hat{L}^2}{r^3} \rangle + \gamma \langle \frac{1}{r^2} \rangle \right)$$

$$\langle \frac{1}{r^3} \rangle = -\frac{\gamma}{l(l+1)} \langle \frac{1}{r^2} \rangle = \frac{\gamma^3}{n^3 l(l+1)(l+1/2)}$$

10. a) $|S=1/2, S_z=\pm 1/2\rangle \quad |n_+\rangle = \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{-i\varphi} \end{pmatrix} \quad |n_-\rangle = \begin{pmatrix} -\sin \frac{\theta}{2} \\ \cos \frac{\theta}{2} e^{i\varphi} \end{pmatrix}$

$$\hat{J}_+(\bar{n}) = |n_+\rangle \langle n_+| = \begin{pmatrix} p_{++} & p_{+-} \\ p_{-+} & p_{--} \end{pmatrix}$$

$$p_{++} = \langle + | n_+ \rangle \langle n_+ | + \rangle = \cos^2 \varphi$$

$$p_{--} = \sin^2 \theta / 2$$

$$p_{+-} = \cos \frac{\theta}{2} \sin \frac{\theta}{2} e^{-i\varphi} \quad p_{-+} = \cos \frac{\theta}{2} \sin \frac{\theta}{2} e^{i\varphi}$$

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$$p_{+}(\bar{n}) = \frac{1}{2} \left(1 + \cos \varphi \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + \sin \theta \cos \varphi \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \sin \theta \sin \varphi \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \right) =$$

$$= \frac{1}{2} (\hat{1} + (\bar{n}, \vec{\sigma})) \quad , \quad \text{где } \vec{\sigma} - \text{антенны}$$

$$b) \quad W(S_0=1/2 | S_e=1/2) = \text{tr}(\hat{J}_+(\bar{B}) \hat{J}_+(\bar{a})) = \frac{1}{4} \text{tr}((1+a_z \sigma_z)(1+b_z \sigma_z)) = \\ = \frac{1}{4} \text{tr}(\hat{1} + a_z b_z \sigma_z \sigma_z) = \frac{1}{2} + \frac{a_z b_z}{4} \text{tr}(\sigma_z \sigma_z) = \frac{1}{2} (1 + (\bar{a}, \bar{b}))$$

11. $\text{tr}(\hat{p}_1) \neq 1 \Rightarrow \text{не сбн. м.н.}$

$$2) \rho_2 \neq \rho_2^+, \text{tr}(\rho_2) \neq 1 \Rightarrow \text{re sbn. m.n.}$$

$$3) \rho_3 \neq \rho_3^+ \Rightarrow \text{re sbn. m.n.}$$

$$4) \text{tr}(\rho_4) = 1, \rho_4 = \rho_4^+, \lambda_i \geq 0 \Rightarrow \text{sbn. m.n.}$$

$$5) \text{kp. Cumbecoppe} \Rightarrow \text{re non. onp.} \Rightarrow \text{c.z. reecopp.} \Rightarrow \text{re sbn. m.n.}$$

$$\underline{13.} \quad |S=1, S_z\rangle \Rightarrow \hat{S} = \frac{1}{4} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix}$$

$$|1,1\rangle = |+\rangle, |1,0\rangle = |0\rangle, |1,-1\rangle = |-\rangle$$

$$S_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad S_y = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}$$

$$S_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$|+\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ \frac{1}{\sqrt{2}} \\ 1 \end{pmatrix}$$

$$|+\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ i/\sqrt{2} \\ 1 \end{pmatrix}$$

$$|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$$|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$$|-\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ -1/\sqrt{2} \\ 1 \end{pmatrix}$$

$$|-\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ -i/\sqrt{2} \\ 1 \end{pmatrix}$$

$$w(a_n | p) = \text{tr}(\hat{P}_{a_n} \hat{p}) = \langle a_n | \hat{p} | a_n \rangle$$

$$\hat{p}' = \sum_i w(a_i | p) \hat{P}_{a_i}$$

$$\hat{p}'_x = \sum_{S_x=\pm 1, 0} w(S_x | p) \hat{P}_{S_x}$$

$$\hat{p}'_+ = \frac{1}{3} (\hat{p}'_x + \hat{p}'_y + \hat{p}'_z) = \frac{1}{48} \begin{pmatrix} 15 & 2i & 6 \\ -2i & 14 & 2i \\ 6 & -2i & 19 \end{pmatrix}$$

$$w(S_y \neq 0 | p_+) = 1 - w(S_y = 0 | p_+) = \frac{23}{48}$$

$$w(S_y = 0 | p_+) = \text{tr}(\hat{P}_{0y} \hat{p}_+) = \frac{25}{48}$$

$$\langle S_x \rangle = \text{tr}(p_+ S_x) = 0$$

$$\underline{14.} \quad |\psi(0)\rangle = |n_+\rangle = \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{i\varphi} \end{pmatrix}$$

$$\hat{\rho}(0) = |n_+\rangle \langle n_+| = \frac{1}{2} (1 + (\hat{b}_1 \bar{n})) \quad \hat{H} = \frac{\omega_L}{2} \hat{b}_3$$

$$\hat{U}(t) = e^{-i\hat{H}t} = e^{-i\frac{\omega_L}{2} \hat{b}_3 t} = \begin{pmatrix} e^{-i\omega_L t/2} & 0 \\ 0 & e^{i\omega_L t/2} \end{pmatrix}$$

$$|\psi(t)\rangle = \hat{U}(t) |\psi(0)\rangle$$

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$$|\psi(t)\rangle = \begin{pmatrix} e^{-i\omega_L t/2} \cos \frac{\theta}{2} \\ e^{i\omega_L t/2} \sin \frac{\theta}{2} e^{i\varphi} \end{pmatrix} \Rightarrow$$

$$\Rightarrow \hat{\rho}(t) = |\psi(t)\rangle \langle \psi(t)| = \begin{pmatrix} \cos^2 \frac{\theta}{2} & \frac{1}{2} \sin \theta e^{-i(\varphi + \omega_L t)} \\ \frac{1}{2} \sin \theta e^{i(\varphi + \omega_L t)} & \sin^2 \frac{\theta}{2} \end{pmatrix}$$

$$\underline{\text{Задача}} \quad |+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\text{Сгенерируем оператор } y: R_y(\theta) = e^{-i\frac{\theta}{2} \hat{b}_y} \Rightarrow$$

$$\Rightarrow R_y(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix} \Rightarrow R_y(\theta) |+\rangle = \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} \end{pmatrix} \Rightarrow$$

$$\Rightarrow \cos \frac{\theta}{2} = \frac{1}{\sqrt{2}} = \sin \frac{\theta}{2} \Rightarrow \theta = \frac{\pi}{2} \Rightarrow \text{т.е. } \frac{\pi}{2} \text{ для } y.$$

$$\text{В квант. механике } \langle E | E' \rangle = \delta(E - E') \Rightarrow \langle E | E \rangle = 1$$

$$\text{т.е. нормировка} \Rightarrow \frac{\partial \langle E \rangle}{\partial x} \text{ т.е. энр.}$$

Также E - квант. $\Rightarrow E$ т.е. энерг. ур. от $\hbar$, просто конст.

